

Collision Warning with Full Auto Brake and Pedestrian Detection - a practical example of Automatic Emergency Braking

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Abstract—More and more vehicles are being equipped with Automatic Emergency Braking (AEB) systems. These systems intend to help the driver avoid or mitigate accidents by automatically applying the brakes prior to an accident. Initially only rear-end collision were addressed but over time more accident types are incorporated and brakes are applied earlier and stronger, in order to increase the velocity reduction before the accident occurs. This paper describes one of the latest AEB systems called Collision Warning with Full Auto Brake and Pedestrian Detection (CWAB-PD). It helps the driver with avoiding both rear-end and pedestrian accidents by providing a warning and, if necessary, automatic braking using full braking power. A limited set of accident scenarios is selected to illustrate the theoretical and practical performance of this system. It is shown that the CWAB-PD system can avoid accidents up to 35 km/h and can mitigate accidents achieving an impact speed reduction of 35 km/h. To the best of the authors knowledge CWAB-PD is the only system on the market that automatically can avoid accidents with pedestrians.

I. INTRODUCTION

AUTOMAKERS are increasingly focusing their safety efforts to the area of active safety. Traditionally the efforts were focused on protecting vehicle occupants and vulnerable road users (VRU's) in case of an accidents, so-called passive safety, but nowadays automakers try to help drivers in avoiding the accident from occurring in the first place. Automakers use active safety technologies that "can detect hazardous traffic situations and actively assist road users in avoiding or mitigating accidents". [1]. The practical application of these technologies have become possible due to recent advances in the area of exterior sensors such as radar, lidar and vision, in the combination with continuous costs reductions. These advances have opened up many possibilities for customer features like lane departure warning (LDW), forward collision warning (FCW) and adaptive cruise control (ACC). In this paper a particular safety feature will be discussed that is called Collision Warning with Full Auto Brake and Pedestrian Detection (CWAB-PD). This features, as launched on the 2011 MY Volvo S60, combines an FCW function with an Automatic Emergency Braking (AEB) function. The objective of this paper is to describe and compare the theoretical and practical performance of the automatic emergency braking provided

by CWAB-PD in commonly used test scenarios. Section II describes the CWAB-PD system. In section III the selection of test scenarios will be motivated, while section IV provides a theoretical analysis on what typically can be achieved with these types of systems. In section V the practical results are shown and explained and the paper finishes with a discussion in section VI.

II. CWAB-PD

Collision Warning with Full Auto Brake and Pedestrian Detection is the third generation of the Volvo system that helps drivers to avoid collisions. The first 2 generations [2] provide warning, brake support and partial automatic emergency braking of 5 m/s^2 in rear-end accidents with moving and stationary vehicles. The innovations in the third generation are:

- Full automatic emergency braking (up to 10 m/s^2).
- Warning, brake support and automatic emergency braking in pedestrian accidents (refer Fig. 1).
- Automatic collision avoidance.



Fig. 1. Visualization of Pedestrian Detection.

The system uses a combination of a long-range radar and a forward-sensing wide-angle camera to continuously monitor the area in front of the vehicle.

- A 640 by 480 pixel black and white progressive scan CMOS forward looking camera (FLC), which is mounted behind the windscreen, is used mainly for classifying the objects, *e.g.* as vehicles or pedestrians, in a 48° field of view (FOV). Since the FLC is used for reporting both vision objects and lane markings, the field of view was chosen to work

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for both.

- A 77 GHz electronically scanning forward looking radar (FLR), which is mounted in the vehicle grille, measures target object information in front of the vehicle in a 60° field of view (FOV), *e.g.* range, range rate and azimuth angle.



Fig. 2. From top-left: Forward Looking Camera (FLC) mounted behind windscreen, Forward Looking Radar (FLR) installation behind grille and the control unit, forward sensing module (FSM).

Data from these two sensors is fused in a separate control unit, Forward Sensing Module (FSM). Fusing data from two sensors with different characteristics reduces the risk of detecting false targets and increases the confidence and data accuracy of the detected target. This is important to allow for hard and early braking in accidents scenarios while simultaneously keeping the risk of false brake interventions at an acceptably low level.

CWAB-PD will provide a warning and brake support when there is a credible risk for an accident. Additionally, if the driver does not intervene in spite of the warning and the possible collision is about 1 seconds ahead, *i.e.* the collision threat becomes imminent, intervention braking is automatically applied to slow down the car. In this way it may not always be possible to avoid a collision, but the impact speed can be reduced significantly, and in many low-speed scenarios an accident can be avoided completely.

The CWAB-PD system is optional and works in parallel to City Safety [3]. The brake system executes the maximum brake request of the two features in case these are not the same.



Fig. 3. Collision Warning when there is an accident risk.

III. TEST SCENARIOS

A. Positive performance

The objective of AEB systems is to automatically avoid or mitigate an accidents in case a driver does not react in time. Most systems that are currently on the market only provide mitigation in rear-end collisions, but in principle AEB can be applied to all collision types. Mitigation is obtained through decreasing the collision impact speed and thereby the risk of injury for the persons involved. In special cases a speed reduction can avoid a collision completely, *e.g.* in a following situation where the potential speed reduction is larger than the relative speed.

The positive performance of a safety feature refers to performance in case of an accident. To understand what this means one has to look at real-world accident statistics. The development of automotive safety systems at Volvo Cars is based on the so-called "Circle of Life" [4]. This working process starts with analyzing real-world accident statistics to understand the types of collisions, their frequencies and their causes. In the real-world accidents vary in terms of velocities, types of vehicles involved, friction, lighting conditions *etc.* This information is crucial for selecting the sensors, their field-of-view, brake performance *etc.* Also understanding the common causes of an accidents is valuable knowledge when designing the human-machine interface that needs to alert the driver when there is an accident risk.

Based on this analysis the positive performance can be predicted, but the actual real-world positive performance can only be estimated once the feature has been on the market for a long period of time. Alternatively, the performance can be estimated using a selected number of scenarios, similar as is being done for passive safety in Euro-NCAP or US-NCAP tests. But, contrary to passive safety there are no well-established and widely-applied test and rating scenarios for AEB systems.

For some active safety features there are some scenarios formally described. *E.g.* NHTSA in the US has rating methods for Electronic Stability Control (ESC), Lane Departure Warning (LDW) [5] and Forward Collision Warning (FCW) [6] and MLIT in Japan has developed guidelines for several active safety features.

Also for AEB systems, such as CWAB-PD, there are some common test scenarios in which positive performance is estimated:

NHTSA

NHTSA [6] has proposed three rear-end collision scenarios

for FCW that can be used for evaluating AEB as well. These scenarios are based in the crash frequency, cost and harm data from GES accident statistics:

- Approach a stationary lead vehicle in 72.4 km/h (45 miles/h)
- Follow a lead vehicle in 72.4 km/h at a constant headway distance of 30 m, where the lead vehicle suddenly brakes with constant deceleration of 3 m/s^2 (0.3 g) within 1.5 seconds.
- Approach a slower lead vehicle in 72.4 km/h. The lead vehicle is moving with a constant speed of 32.3 km/h (20 miles/h).

Tatcham

In [7] motivates the importance of low-speed collisions, based on statistics from insurance companies. The test scenario that is used in this publication is:

- Approach a stationary lead vehicle in 16 km/h.

MLIT

MLIT has a number of requirements on AEB systems. Several are related to the risk of misuse of the system. One of these is that the stopping distance of a collision avoidance system has to be *less* than one meter.

Auto Motor und Sport

Also the automotive press shows an increasing interest in AEB system. One of the first tests was performed by the German magazine Auto Motor und Sport [8] and vehicles from different manufacturers were compared in the following scenarios:

- Approach a stationary lead vehicle in 25 km/h.
- Approach a stationary lead vehicle in 50 km/h.
- Follow a lead vehicle in 50 km/h at a constant headway of 15 m, after which the lead vehicle suddenly brakes maximally.

The scenarios above are all directed towards rear-end collisions. For pedestrian accidents there are, to the best of the authors knowledge, no common test scenarios used for AEB systems. There is an increased focus on pedestrian or Vulnerable Road User (VRU) safety, but most technology is directed towards passive safety systems. Statistics show that, in Sweden, about 16 % of all traffic fatalities are pedestrians [9] and in the USA it is about 11 % of all traffic fatalities (4,700 people) [10]. It is also known that about 50 % of the pedestrian accidents occur in speeds below 25 km/h [11].

EuroNCAP carries out a series of passive safety tests to replicate accidents involving child and adult pedestrians, where impacts occur at 40kph (25mph). Impact sites are then assessed and rated fair, weak and poor. This scenario can also be used for testing CWAB-PD.

B. Negative performance

Besides the positive performance of AEB systems, *i.e.* the performance in collision situations, one has to address negative performance as well. The risk of automatic emergency braking has to be minimized in non-collision situations in order to not disturb the driver under normal driving situations. These tests are not in the scope of this paper, but the CWAB-PD system has been extensively tested on public roads in different traffic environments, in different countries and by different drivers and it has been confirmed with sufficient confidence that the risk for false interventions is acceptably low.

IV. THE THEORETICAL POTENTIAL OF AEB SYSTEMS

To calculate the potential of an AEB system one needs to assume a certain prediction model. Automatic braking has to be applied a certain time before a collision occurs, so one has to estimate how the road users involved will behave during a certain prediction horizon. There are several models that can be used for this, e.g. assume constant acceleration or constant turn radius. Examples are provided in [12], [13] and [14].

A common measure for calculating the collision risk is time to collision (TTC). Assuming two objects move in the same direction and assuming constant acceleration, the time to collision can be expressed as:

$$t_{\text{collision}} = \begin{cases} -\frac{\tilde{p}_x}{\tilde{v}_x}, & \tilde{v}_x < 0 \text{ and } \tilde{a}_x = 0 \\ -\frac{\tilde{v}_x}{\tilde{a}_x} - \frac{\sqrt{\tilde{v}_x^2 - 2\tilde{p}_x\tilde{a}_x}}{\tilde{a}_x}, & \tilde{v}_x < 0 \text{ and } \tilde{a}_x \neq 0 \\ -\frac{\tilde{v}_x}{\tilde{a}_x} + \frac{\sqrt{\tilde{v}_x^2 - 2\tilde{p}_x\tilde{a}_x}}{\tilde{a}_x}, & \tilde{v}_x \geq 0 \text{ and } \tilde{a}_x < 0 \\ \text{undefined} & \tilde{v}_x \geq 0 \text{ and } \tilde{a}_x \geq 0 \\ \text{undefined} & \tilde{v}_x^2 - 2\tilde{p}_x\tilde{a}_x < 0 \end{cases}$$

Where $\tilde{p}_x$, $\tilde{v}_x$ and $\tilde{a}_x$ are the relative position, velocity and acceleration, respectively. So, when the host vehicle is approaching a stationary target object with constant velocity, the time to collision is linearly decreasing with the relative position according:

$$t_{\text{collision}} = -\frac{\tilde{p}_x}{\tilde{v}_x} = -\frac{\tilde{p}_x}{v_{x,\text{host}}}, \quad v_{x,\text{target}} = 0$$

When approaching a target the driver has can steer around the object, brake before the object or perform a combination of the two.

The time needed to avoid collision by braking can be expressed as:

$$t_{\text{brake}} = -\frac{\tilde{v}_x}{a_{x,\text{host}}}$$

This is the time needed to reduce the relative velocity to zero. The distance that the host will travel during the braking maneuver is:

$$\tilde{p}_{x,brake} = -\frac{a_{x,host} t_{brake}^2}{2} = -\frac{\tilde{v}_{x,0}^2}{2a_{x,host}}$$

and thus the time to collision at which braking has to be commenced is:

$$ttc_{brake} = -\frac{\tilde{p}_{x,brake}}{\tilde{v}_{x,0}}$$

Note that this is the time to collision given that no braking intervention occur. If braking occurs, then the time to collision will be higher, and when collision avoidance is achieved, TTC approaches infinity. The time needed to steer a vehicle around a target can be expressed as:

$$ttc_{steer} = \min \left(\sqrt{\frac{2}{a_{y,host}} \left(y_0 \pm \frac{w_{host} + w_{target}}{2} \right)} \right)$$

where w_{host} and w_{target} is the width of the host and the target, and $a_{y,host}$ is the expected maximal lateral acceleration of the host vehicle. Both ttc_{brake} and ttc_{steer} can be plotted as a function of the relative velocity, resulting in Fig. 4.

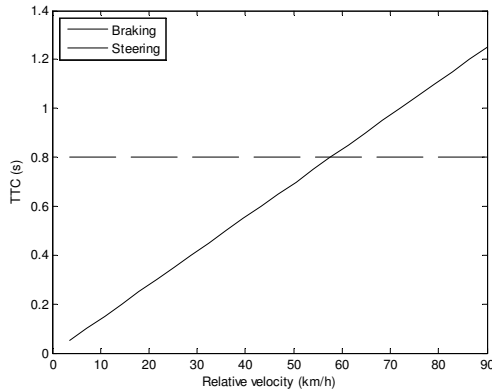


Fig. 4. The time to collision where the accident becomes unavoidable by braking or steering. A conservative AEB system can only intervene below both these curves.

However, the assumptions in this prediction model are very conservative and therefore lead to unnecessarily late braking, especially in low-speed situations. The model has therefore been extended by incorporating:

- Brake system dynamics. A deceleration cannot be achieved instantaneously as the brake system needs to build up brake torque first.
- Stop distance. When targeting collision avoidance one needs to incorporate a certain between the target and host vehicle.
- Vehicle dynamics. At lower speeds one cannot generate large later acceleration.

These phenomena have been incorporated in the CWAB-PD system, the diagram is shown in Fig. .

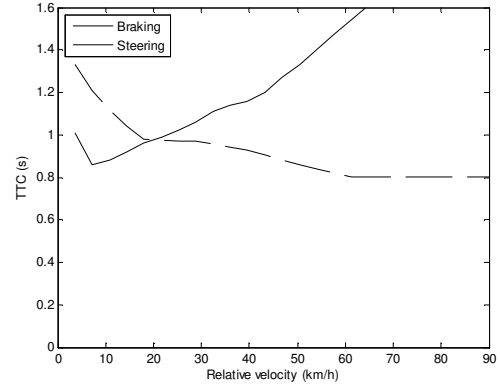


Fig. 5. The time to collision where the accident becomes unavoidable by braking or steering using the extended model in CWAB-PD.

Note: Even further advances have been made in [16], but these have not yet tested in practice.

Automatic emergency braking is applied when the driver cannot brake or steer anymore to avoid a collision. The CWAB-PD applies maximum braking power resulting in decelerations of up to 10 m/s^2 .

The theoretic speed reduction that is achieved based on the TTC curve in Fig. 5 can be obtained using a simple brake model. The brake model that is used here consist of a pure delay of 180 ms and then a deceleration ramp of 20 m/s^3 . The assumed friction coefficient is set to one. This gives the expected velocity reduction is shown in Fig. 6.

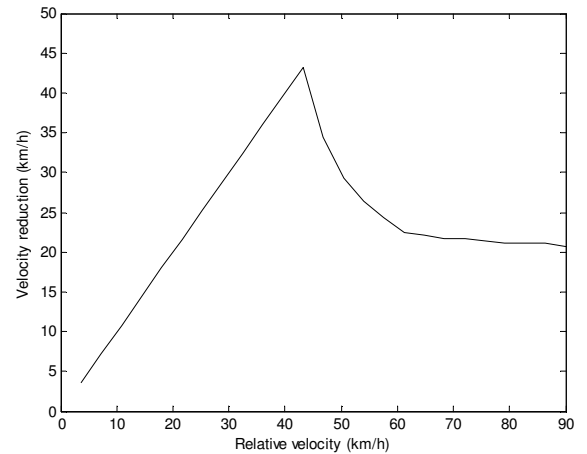


Fig. 6. The theoretical speed reduction based on the ttc curves in Fig. 4 and a simple brake model.

V. PRACTICAL RESULTS

In section III.A a number of scenarios were selected for testing AEB systems. The CWAB-PD was tested in these scenarios and the results are illustrated below. The first test show a scenario where the host vehicle approach a stationary pedestrian, see Fig. 7, which also show the test setup and a snapshot of the vision system. Fig. 8 then shows the achieved speed reduction which is compared to the theoretical curve in section IV.



Fig. 7. The test setup used for evaluating pedestrian performance of the CWAB-PD system.

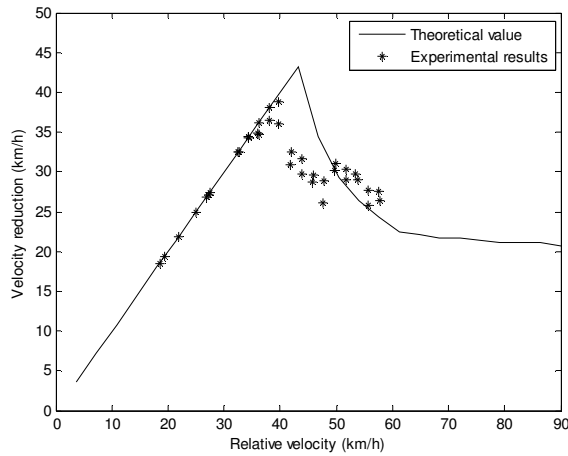


Fig. 8. The theoretical speed reduction together with real life performance. The difference between theoretic and experimental results is mainly due to the simple brake model used in the theoretical curve.

NHTSA

In the NHTSA test case a) where a stationary lead vehicle is approached in 72.4 km/h the achieved speed reduction of the CWAB-PD system is 25 km/h giving a relative impact speed of 47 km/h. Test case b) has not been run due to limitations in the test equipment. In test case c), approach a slower lead vehicle in 72.4 km/h, where the lead vehicle is moving with a constant speed of 32.3 km/h, the system reduces the impact speed with 35 km/h, giving a relative impact speed of 5 km/h.

MLIT

MLIT has a number of requirements on AEB systems. Several are related to the risk of misuse of the system. One of these is that the stopping distance of a collision avoidance system has to be *less* than one meter. The performance in this respect of the CWAB-PD system is shown in Fig. 9.

Thatcham

In the scenario where a stationary lead vehicle is approached at 16 km/h, the CWAB-PD system achieves collision avoidance.

Auto Motor und Sport

When approaching a stationary lead vehicle at 25 km/h, the CWAB-PD system achieves collision avoidance. When approaching a stationary lead vehicle at 50 km/h, the speed reduction is approximately 30 km/h. For two vehicles traveling 15 m apart at 50 km/h and the leading vehicle suddenly brake maximally, the CWAB-PD system is also able to reduce the impact speed by 30 km/h.

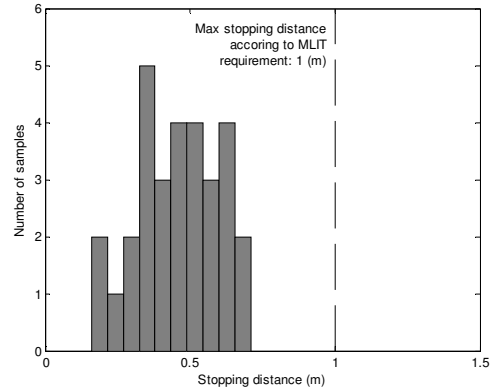


Fig. 9. The stopping distance for a number of pedestrian scenarios of speeds between 5 – 35 km/h. One of the MLIT requirements is that a collision avoidance system has to stop closer to the obstacle than 1 m.

VI. DISCUSSION

Automatic Emergency Braking is an effective way to decrease the impact speed in rear-end and pedestrian accidents. The Volvo CWAB-PD system is, to the best knowledge of the authors, the only system on the market that automatically can avoid these accidents in vehicle speeds up to 35 km/h. The actual speed reduction is dependent on several factors such as road conditions, tire conditions, traffic situation etc. The system shows very good performance in common test scenario from NHTSA, Thatcham and Auto Motor and Sport.

From research it is known that a reduction of speed before impact with 10 % can reduce fatal injuries in car crashes with approximately 30 % [17] and that lowering the impact speed in pedestrian collisions from 50 to 25 km/h reduces fatality risk by 85 % [11]. This illustrates what the real-life benefit of the CWAB-PD system may be.

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